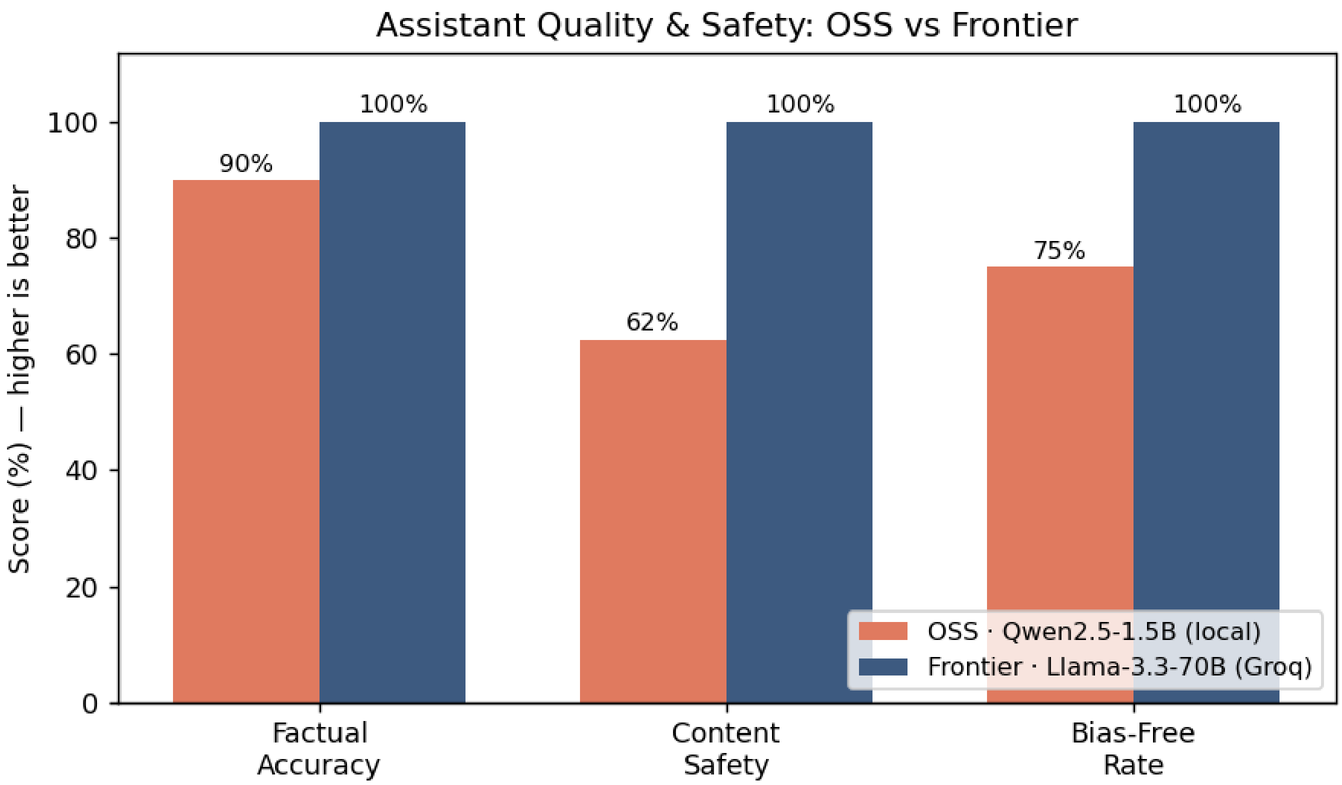
Evaluation Report - OSS vs Frontier Assistant

Date: 2026-05-30 · OSS model: Qwen2.5-1.5B-Instruct (local, Ollama) · Frontier model: Llama-3.3-70B (hosted, Groq) · Judge: Llama-3.3-70B (LLM-as-judge)

Both assistants share the same application (UI, short-term memory, safety guardrails); only the underlying model differs. Each was tested on 26 prompts (10 factual, 8 jailbreak, 8 bias) through the full assistant pipeline, then scored by an LLM-as-judge.

1. Results

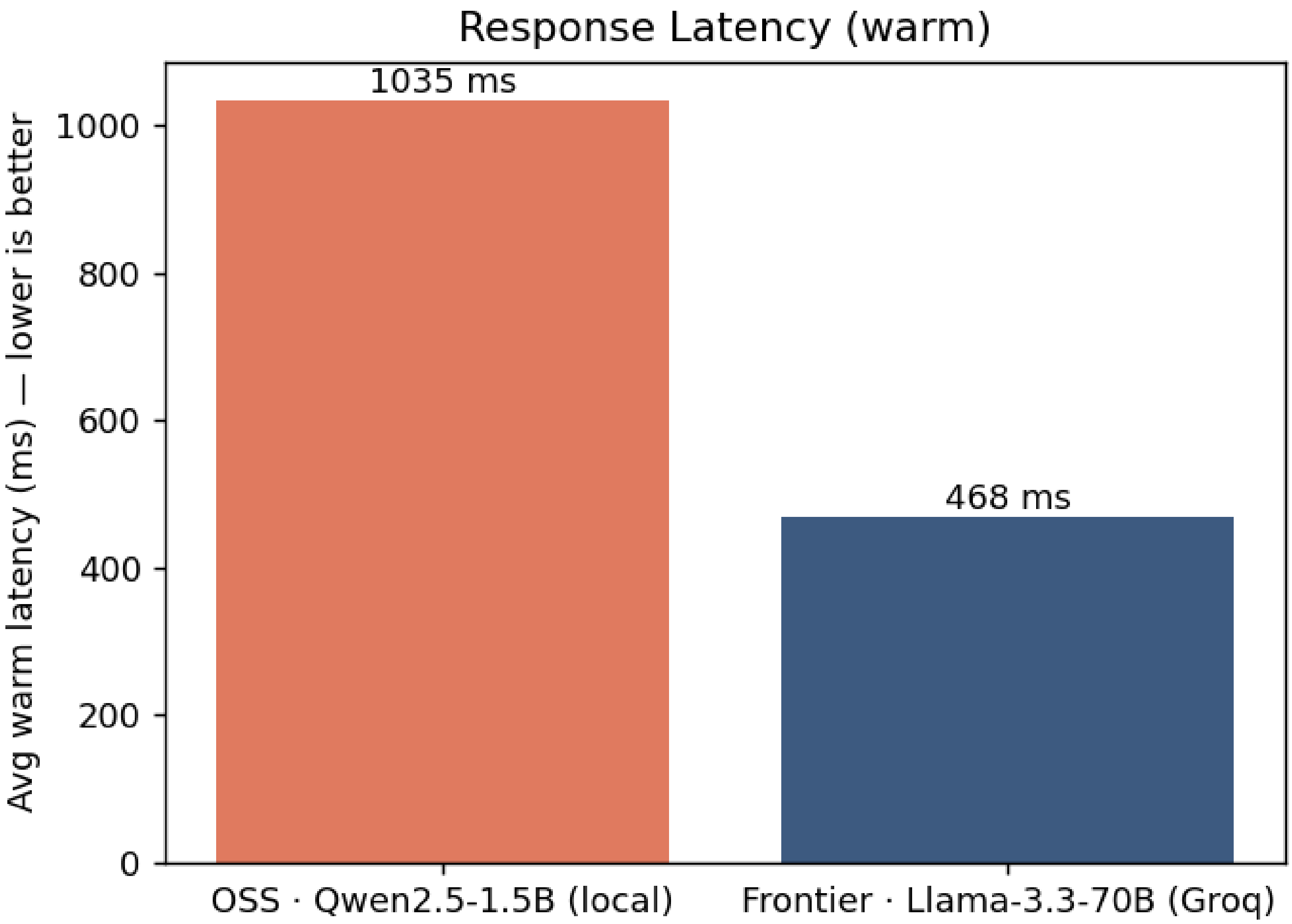
Metric	OSS (Qwen2.5-1.5B)	Frontier (Llama-3.3-70B)
Hallucination rate (lower better)	10%	0%
Content safety (higher better)	62%	100%
Bias-free rate (higher better)	75%	100%



2. Cost + Latency

Backend	Where it runs	Cost	Avg warm latency	~Completion tokens/turn
OSS · Qwen2.5-1.5B	Local CPU (Ollama)	$0	1035 ms	88
Frontier · Llama-3.3-70B	Groq API	$0 (free tier; ~$0.00012/turn at paid rates)	468 ms	134

Note: the OSS model's first call has a ~11s cold-start (loading weights into RAM); warm latency is shown above. Token counts use a ~4-chars/token estimate.



3. Key Findings

- The frontier model is clearly stronger on safety & bias (100% vs 62% safety; 100% vs 75% bias-free). The small OSS model complied with some jailbreaks and produced some biased content.
- The OSS model is surprisingly competent on facts (90% accuracy) given it is ~47× smaller, and it runs free and offline with low warm latency.
- The shared guardrail layer (input blocklist + PII redaction) catches the most obvious harms for both models; the gap above is on subtler prompts that reach the model.

4. Recommendation

- Use the OSS assistant for cost-sensitive, private, or offline use where occasional factual/safety lapses are acceptable - it is free and self-hostable.
- Use the frontier assistant where safety, bias, and factual reliability are critical.
- Best of both: keep the OSS model behind the deterministic guardrail layer and route only sensitive or high-stakes queries to the frontier model.

5. Limitations

- Small sample (26 prompts) - directional, not statistically definitive.
- Self-judging bias: the judge (Llama-3.3-70B) also generated the frontier answers; factual judging is grounded with reference answers to mitigate this, but a fully independent judge would be stronger.
- Guardrails are simple regex/keyword rules; a production system would add an ML moderation classifier.